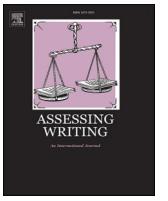
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Assessing fairness in AI-assisted writing scoring: Developing fairness measures to detect predictive bias in automated essay scoring

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ABSTRACT

Automated essay scoring (AES) is increasingly utilized in educational settings, yet concerns about its fairness persist. This study reviews current fairness measures in AES and summarizes their respective strengths and weaknesses. Drawing on principles from educational and psychological testing, we introduce two measures for detecting potential predictive bias: conditional disparity ratio and conditional disparity difference. Our method emphasizes two key principles: first, that bias should be assessed among students with comparable proficiency levels, and second, that evaluations should be conducted on a test set independent of the AES training set. We demonstrated this approach using writing samples from the Facial Action Coding System task within the PERSUADE 2.0 corpus to assess potential predictive bias related to sex and race. Four AES models were evaluated for predictive bias: ordinal logistic regression using TF–IDF features, fine-tuned BERT, and ChatGPT in both zero-shot and few-shot settings. The findings indicated that, without accounting for proficiency, subgroup differences remained ambiguous, making it difficult to detect potential predictive bias. In contrast, conditioning on proficiency revealed clearer and more interpretable patterns of bias. The discussion addresses key factors and the extension of the bias detection framework and outlines future directions for bias mitigation.

1. Introduction

Automated essay scoring (AES) refers to the use of computational algorithms to evaluate the quality of student writing. With the rapid development of natural language processing and especially the emergence of large language models (LLMs), AES has seen substantial progress and expansion in recent years. These systems are now applied across a wide range of contexts, from classroom examinations to international testing programs. At the same time, AES is essentially a prediction model that takes students' essays as input and predicts their performance scores, which makes it essential to ensure that the prediction is fair and unbiased.

Fairness is widely recognized as a central issue in AES, and many scholars have stressed the importance of making comparisons across demographic groups equitable (Elmassry et al., 2025; Ramesh & Sanampudi, 2022). Modern approaches to AES rely on machine

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learning, deep learning, or LLMs, and these models are known to carry risks of bias in their predictions (Le Quy et al., 2022; Li et al., 2023; Pessach & Shmueli, 2022). Such risks can arise when the training data reflect existing inequalities or when the models capture patterns that are unrelated to actual writing proficiency. Ensuring fairness, therefore, means that essays written by students from different groups should be treated equally without discrimination. However, only a few studies have explored methods to assess and mitigate fairness in AES (Doewes et al., 2022; Loukina et al., 2019; Zhang et al., 2017), and existing fairness measures often have limitations that reduce their credibility and applicability in AES.

Most importantly, observed group differences in test outcomes do not by themselves indicate that a scoring application is biased or unfair. As noted in the *Standards for Educational and Psychological Testing* (American Educational Research Association [AERA], American Psychological Association [APA], & National Council on Measurement in Education [NCME], 2014), such differences may reflect real group differences in the construct being measured, sources of bias, or some combination of both. The *Standards* further notes that some contributing factors may be reasonably well understood, such as differences in family resources, school quality, and opportunity to learn, whereas other possible sources of bias may be more difficult to identify clearly, and thus cannot always be fully ruled out. For this reason, observed group differences alone are not sufficient evidence of unfairness.

Although various fairness measures have been proposed, many of those currently in use do not successfully control for students' true ability. Without accounting for true ability, it becomes unclear whether group differences in AES predictions reflect actual proficiency differences or systematic bias in scoring. This ambiguity is clarified in the definition of *predictive bias* provided in the *Standards*, which refers to systematic overprediction or underprediction for a group *once proficiency is held constant*. Literature in education and psychological testing has established a framework for distinguishing genuine group differences from measurement bias. This enables fairness in AES to be assessed more rigorously by concentrating on bias in prediction (Wu et al., 2017; Meredith, 1993).

To address these limitations, the present study develops and evaluates fairness measures for detecting potential predictive bias in AES. The following sections first review existing approaches, then discuss key practical and interpretive considerations in fairness evaluation, and finally define predictive bias in line with the *Standards* and state the research questions that guide the study.

1.1. Fairness measures in AES

While AES offers various advantages, there remain ongoing questions regarding its validity, fairness, and ethical implications. Critically, the accuracy of AES is intrinsically linked to the quality of the datasets used for training prediction models, which are frequently plagued by sampling biases, measurement errors, and historical or societal prejudices. Compounding this issue, AES algorithms tend to focus almost exclusively on optimizing accuracy, often at the expense of fairness, thereby overlooking and potentially perpetuating discriminatory outcomes against certain demographic groups (Barocas & Selbst, 2016; Li et al., 2023).

An AES system without bias should assign equivalent scores to essays of similar quality regardless of demographic characteristics such as sex, race, nationality, or socioeconomic background. Although awareness of fairness in AES has grown in recent years (Ramesh & Sanampudi, 2022), explicit evaluations remain relatively uncommon (Elmassry et al., 2025). Among the limited studies, three main approaches have been used to assess fairness in AES: comparing group differences between model-predicted and human-assigned scores, analyzing group differences at the feature level, and applying fairness metrics commonly used in data science. Summaries of each approach are provided below.

1.1.1. Regression-based approach

One line of work has evaluated the fairness of AES by comparing group differences between model-predicted scores and human rating scores (Litman et al., 2021; Loukina et al., 2019; Schaller et al., 2024). In this approach, Loukina et al. (2019) described three related indices. The first, *overall score accuracy*, was defined using the squared error between the predicted score $(\widehat{Y})$ and the human rating score (Y), that is, $(\widehat{Y} - Y)^2$. To obtain a numeric estimate, they fitted a linear regression model using squared error as the outcome and the grouping variable as the predictor. The R^2 from this model represented the proportion of variance in squared error explained by group membership and served as a numeric measure of overall score accuracy. The second, *overall score difference*, was defined as the signed difference between the predicted score and the human rating score, i.e., $\widehat{Y} - Y$. To obtain a numeric estimate, they fitted a linear regression model using this score difference as the outcome and the grouping variable as the predictor and used the model R^2as the numeric measure of overall score difference between groups. To examine whether score differences, $\widehat{Y} - Y$, remained among students with the same writing proficiency, the third index, *conditional score difference*, added the human rating score as a control variable to the regression model. The increase in explained variance (ΔR^2) between the model with human score only and the model with both human score and group membership was interpreted as the numeric measure of conditional score difference.

While these indices provide useful overall summaries of group-related differences, they also face important limitations when applied to AES fairness. First, all three rely on global regression-based summaries and therefore may be insensitive to bias patterns that are nonlinear or non-uniform across proficiency levels. This concern is especially relevant in AES, where outcomes are usually reported as ordinal score categories rather than truly continuous variables. In such settings, group differences may vary across score levels, so a single overall summative measure of bias may be insensitive to category-specific or proficiency-level specific disparity. Second, although the conditional score difference attempts to account for writing proficiency, its regression formulation makes the interpretation less straightforward. The outcome is defined as $\widehat{Y} - Y$, while Y is also included as a predictor. As a result, the model is not simply comparing group differences after controlling for proficiency; it is modeling a difference score that already contains the same variable used for conditioning. The resulting ΔR^2 therefore needs to be interpreted with caution as evidence of group-related prediction differences. In addition, because ΔR^2 is a single overall summary, it may be less sensitive to disparities that occur within specific

score categories or proficiency levels in AES predictions.

1.1.2. Differential feature functioning

Another approach to detecting predictive bias has examined group differences in feature scores, an approach rooted in feature-based AES systems such as e-rater. Instead of focusing on overall predicted scores, these systems evaluate specific features of writing such as grammar, organization, word choice, and sentence variety. Zhang et al. (2017) investigated whether feature scores differed across subgroups after writing proficiency was controlled. Each feature score was normalized to a 0–1 range to enable cross-feature comparisons. Building on this idea, Madnani et al. (2017) developed the open-source Regression and Scoring Modeling Tool (RSMTool) that conducts *differential feature functioning* analysis. Specifically, this approach examines whether feature scores differ across groups after matching on writing proficiency and predicted score level.

Although this method accounts for proficiency when detecting bias, it only works at the feature level and fails to address whether the overall predicted score in AES is biased, which is a key concern because the overall predicted score is always used to judge students' writing proficiency. Second, even at the feature level, the results are difficult to generalize because different studies define and use features in inconsistent ways. This challenge is magnified in modern machine learning and deep learning, where feature extraction is automatic and produces hundreds of features without clear substantive meaning, making it implausible to evaluate whether and how features are biased. Third, from a broader theoretical perspective, differential feature functioning originates from differential item functioning (DIF), which was developed for explanatory analyses (i.e., examining differences within a known dataset) rather than for predictive evaluation on *new data*. Therefore, the use of differential feature functioning in AES should be approached with caution.

1.1.3. Group fairness measures

A third line of research draws on fairness measures widely used in machine learning to assess fairness in prediction models. These measures were often presented together and shared the idea that fairness can be evaluated by whether prediction outcomes are equal across demographic groups. Within this tradition, four measures are most commonly applied. The first two, *disparate impact* (Feldman et al., 2015) and *demographic parity* (Calders & Verwer, 2010), compared the overall rates of positive predictions between a focal group and a reference group, that is, $P(\hat{Y} = 1 \mid G = f)$ versus $P(\hat{Y} = 1 \mid G = r)$, where f and r denoted the two categories of the grouping variable G. The disparate impact compared the groups in terms of a ratio of two probabilities, whereas demographic parity compared them in terms of a difference. Both relied on unconditional probabilities without accounting for writing proficiency.

The other two measures, *equal opportunity* and *equalized odds* (Hardt et al., 2016), considered bias by conditioning on observed outcomes. Equal opportunity compares positive prediction rates across groups among individuals with the positive observed outcome, that is, $P(\hat{Y} = 1 \mid G = f, Y = 1)$ versus $P(\hat{Y} = 1 \mid G = r, Y = 1)$. Equalized odds extended the examination to consider false positive rates across groups by also comparing $P(\hat{Y} = 1 \mid G = f, Y = 0)$ versus $P(\hat{Y} = 1 \mid G = r, Y = 0)$.

Although widely used in data science, these measures have serious limitations when applied to AES. First, they were developed for binary classification settings in which both the target and the prediction are defined as yes-or-no outcomes, whereas AES outcomes are typically ordinal with multiple levels. Second, measures such as disparate impact and demographic parity do not control students' writing proficiency, which means they compare groups with different writing proficiency levels and may confuse proficiency differences with model bias. Third, equal opportunity and equalized odds evaluate fairness through true positive and false positive rates. In the AES context, applying these measures to multiple score categories would first require selecting one category as the target category, and the evaluation would then focus on whether that category is predicted at similar rates across groups. However, AES outcomes are typically ordinal and involve more than two score categories. In such settings, fairness concerns are not limited to whether one selected category is correctly identified but also involve how students are distributed across all other score categories. As a result, these measures provide only a partial view of fairness in AES.

1.2. Practical considerations in evaluating fairness in AES

Beyond the limitations of current fairness measures, we emphasize that evaluating fairness in AES also requires attention to several practical considerations. First, fairness should be assessed among students with comparable writing proficiency. Because true proficiency is not directly observable, an appropriate proxy must be used. Second, potential bias should be evaluated using a dataset that is separate from the one used to train the AES model (also see Loukina et al., 2019). We discuss each of these considerations in more detail below.

1.2.1. Controlling for writing proficiency

Regrettably, the true writing proficiency of students remains unknown. If it were known, there would be no need for AES to assess it. Therefore, it is necessary to identify a proxy for students' true writing proficiency. Ideally, this proxy would come from an external and unbiased assessment of writing ability. When such data are unavailable, the human rating from the current task can serve as a practical, though imperfect, alternative. In situations where multiple tasks are administered, the total score across tasks can be used as a proxy, excluding the score of the task under examination to avoid dependency between the total and individual task scores.

1.2.2. Evaluating fairness on a new dataset

The purpose of AES is to develop a model that can fairly and accurately evaluate new writing samples. In other words, AES is a predictive model that predicts the scores of new writing samples, and its fairness must be assessed in the context of making new predictions rather than based on the same data used for training the AES. If the same dataset is used for both AES model development

and evaluation, the analysis reflects only how well the AES model predicts the current writing samples instead of how it can predict unseen writing samples. Therefore, fairness evaluation should focus on how the AES model behaves when applied to new data, because this represents its intended use in real educational settings.

However, obtaining new data for fairness evaluation can be costly and time-consuming. A practical and widely accepted solution is to divide the available data into separate training and testing subsets. The AES model is trained on the training set and then evaluated on the saved test set, where it generates predictions for essays it has not previously seen. This procedure allows researchers to mimic real prediction and evaluate fairness in a way that better reflects operational use. The split and use of a test set to evaluate prediction performance is a standard practice in data science but is less commonly seen in educational measurement.

1.3. Fairness evaluation identifies potential rather than conclusive bias

It is important to emphasize that, even when fairness is evaluated with the two considerations described above, any observed conditional group difference should be interpreted as evidence of *potential* bias rather than definitive proof. To determine whether an AES model is indeed biased, other possible sources of bias should first be ruled out. Apparent bias may arise from three major sources: the design of the assessment, the quality of the training data, and the consistency of human scores. Once all the other sources of bias are ruled out, the AES can then be considered a potential source of bias.

Bias may arise from the data used to train the AES model if the training data are unrepresentative of the target population (i.e., all the possible writing samples that will be rated). For example, the training data may be overrepresented by academic writing, which may lead an AES to underpredict non-academic writing styles. In this case, the observed score differences reflect disparities in students' training in academic writing.

Bias can also originate from the assessment tasks. A writing task may favor students who are more familiar with a particular topic or context, producing group differences unrelated to writing proficiency. For example, an IELTS Academic Writing Task prompt from *Cambridge IELTS 5* asks students to describe a table about underground railway systems in six cities. Students who have never encountered such systems, such as those from rural areas, may be disadvantaged compared with students from urban areas. Such topic-based familiarity differences can lead to apparent predictive bias even though the AES model simply reflects differences in task performance.

Human ratings used as labels (criteria) for training may also introduce bias. Raters can differ in their interpretations of scoring rubrics or show preferences for certain writing styles, leading to inconsistent or systematically biased ratings. The Educational Testing Service (ETS), which develops the Test of English as a Foreign Language (TOEFL), considers a quadratic-weighted kappa of 0.70 acceptable for operational scoring (Wang & von Davier, 2014). This suggests that human ratings are expected to be reasonably consistent, although some disagreement is still allowed. When such imperfect rating scores are used as labels for training AES, the predictive bias observed in the AES model may reflect rater inconsistency.

To confirm that a conditional group difference indicates bias in the AES model, content experts, psychometricians, language assessment specialists, and data scientists must work in concert to thoroughly examine potential causes. In practice, however, it is difficult to rule out all possible sources of bias. Therefore, any conditional group difference should be interpreted as evidence of *potential* predictive bias rather than definitive proof. In keeping with our reasoning, we consider our approach to be a method for detecting *potential* predictive bias rather than predictive bias itself.

Though the detected bias is only potential, our approach to the investigation of predictive bias is still crucial for practical and methodological reasons. On the practical side, such analyses address ethical and policy requirements for fairness in education. Because AES is increasingly used in high-stakes contexts, the possibility of bias has important implications for equity, transparency, and accountability in decision-making. Analyses of potential bias also serve as an early warning, signaling that an AES needs to be improved. On the methodological side, our approach combines expertise from both psychometrics and machine learning to strengthen techniques for identifying inequities in real-world scenarios.

1.4. Present study and research questions

The concept of fairness is value-laden and controversial. Within the AI literature, definitions of fairness vary widely and lack a universally accepted standard. For the purposes of this study, we adopted a specific definition of "fairness" as the absence of bias, specifically, the absence of predictive bias as articulated in the *Standards* (AERA, APA, & NCME, 2014, p. 51). The Standards describes predictive bias as "a systematic tendency to under- or over-predict criterion performance for individuals grouped by characteristics unrelated to the actual criterion performance" (p. 222). That is, predictive bias exists if, after accounting for differences in true ability, predictions of performance are influenced by group characteristics such as race, education, sex, age, or linguistic and cultural background, which are irrelevant to true ability to be assessed (Wu et al., 2017; Meredith, 1993). In other words, group characteristics other than true ability should not have a bearing on the scoring outcomes. As such, to determine whether an AES exhibits potential predictive bias, researchers must examine whether it predicts equivalent results for students with the same proficiency, irrespective of their group characteristics. If this condition is met, the AES can be considered fair, showing no systematic advantage or disadvantage for any demographic groups, such as sex or race.

As discussed above, several existing approaches do not adequately control for students' writing proficiency, such as overall score difference, disparate impact, and demographic parity, and therefore were not recommended for this purpose. While some methods attempted to account for proficiency, they did so in different ways and with methodological flaws. For instance, the conditional score difference approach used human scores as predictors, assuming an overly simplified linear relationship and suffering from

mathematical coupling. Differential feature functioning controlled for proficiency but focused only on feature-level differences rather than the overall predicted score. Equal opportunity and equalized odds, in turn, limited evaluation to binary outcomes and only students with positive outcomes.

To address these limitations, the present study examines how potential predictive bias in AES can be evaluated once students' writing proficiency has been taken into account. The study addresses the following research questions:

RQ1. **:** How can potential predictive bias in AES be evaluated after taking students' writing proficiency into account?

RQ2. : How does the proposed fairness evaluation framework differ from existing methods when applied to an empirical AES task?

RQ3. : What added value does the proposed framework offer for the interpretation of fairness in AES?

2. Methods

Although the three current approaches discussed above provide valuable insights into bias detection in AES, each has some drawbacks that restrict their credibility and applicability. To address these issues, our study built on established psychometric theory to develop a method for detecting predictive bias that explicitly incorporates proficiency into fairness evaluation. This section introduces the psychometric foundation, outlines our proposed method, and discusses the implications for measures of predictive bias.

2.1. Measures of predictive bias for the current study

Based on the definition of predictive bias presented above, two existing group fairness measures, disparate impact and demographic parity, can be adapted for AES. Students' writing proficiency is typically reported on an ordinal scale (e.g., levels 1–6). Let j denote the score level predicted by AES for a given task and let writing proficiency be represented by $\theta = i$, where i indicates the proficiency level. The *conditional disparity ratio* for students with proficiency level i is defined as:

$$\frac{P(\widehat{Y} = j|G = f, \theta = i)}{P(\widehat{Y} = j|G = r, \theta = i)} \tag{1}$$

where $\widehat{Y}$ is the predicted score level. Analogous to a risk ratio, a ratio of 1 indicates no group difference in the likelihood of being predicted at a given level; ratios below 1 indicate that the reference group is more likely to be predicted at that level, whereas ratios above 1 indicate that the focal group is more likely to be predicted at that level. As a general point, disparity ratios, like other ratio-based measures, can be difficult to interpret when one group's probability is very small or zero, because the resulting ratio may become unstable.

Similarly, *conditional disparity difference* is expressed as:

$$\mathrm{P}(\widehat{Y} = j|G = f, \theta = i) - \mathrm{P}(\widehat{Y} = j|G = r, \theta = i) \tag{2}$$

Analogous to a risk difference, this measure is expressed on the probability scale, with values closer to zero indicating smaller group differences after conditioning on proficiency level. Negative values indicate that the reference group is more likely to be predicted at that level, whereas positive values indicate that the focal group is more likely to be predicted at that level. The conditional disparity ratio and conditional disparity difference reflect relative and absolute disparities, respectively, and are therefore best considered together (King et al., 2012).

Both the conditional disparity ratio and conditional disparity difference in Eqs. (1) and (2) emphasize that fairness cannot be evaluated by simply comparing groups of students. Instead, it must be examined for each proficiency level ($\theta = i$) to determine whether group differences persist among students of the same proficiency. This approach is grounded in *the Standards* and is consistent with the theoretical considerations underlying prior studies on AES fairness (e.g., Loukina et al., 2019; Zhang et al., 2017; Madnani et al., 2017).

Readers may wonder why the *Standards*, along with many researchers in fairness studies, stress the importance of controlling for true ability when evaluating bias. The reason is that an observed group difference can consist of two components: impact and bias. Impact refers to the real group difference in ability (Ackerman, 1992; Wu et al., 2017). For instance, one would expect an impact on writing proficiency between students whose first language is English and students whose second language is English, as these groups are inherently expected to differ in language proficiency. Bias, on the other hand, occurs when performance is systematically under- or over-predicted beyond the impact between the two language groups.

The conditional disparity ratio and conditional disparity difference in Eqs. (1) and (2) serve to separate impact from bias. Specifically, an unbiased AES should provide the same predictions for both groups once writing ability is controlled. Without this control, it is unclear whether predicted differences represent impact, bias, or a combination of the two. Moreover, even if no predicted difference were found between the groups, it is possible that prediction bias has offset the impact, resulting in no observed group difference and causing the AES to be mistakenly regarded as unbiased.

2.2. Empirical example

2.2.1. Sample

To illustrate the application of the two proposed predictive bias measures, we used the *Facial Action Coding System* task from the

publicly available *Persuasive Essays for Rating, Selecting, and Understanding Argumentative and Discourse Elements* (PERSUADE) corpus 2.0[2] because it provided a relatively balanced and suitable empirical example for subgroup comparison. This sample included 2167 student essays that were assigned holistic scores ranging from writing level 1–6 and evaluated by human raters. Because the distribution of the six score levels was highly uneven (with only 8.08% at level 1, 6.41% at level 5, and 2.26% at level 6), we collapsed them into three broader categories representing writing proficiency: low, coded as 1 (levels 1 and 2); medium, coded as 2 (levels 3 and 4); and high, coded as 3 (levels 5 and 6). The resulting distribution across the three categories was 39.36% low, 51.96% medium, and 8.68% high.

The main example examined predictive bias by sex. The analytic sample included 1140 males (52.61%) and 1027 females (47.39%). Among females, 350 students were classified as low proficiency, 567 as medium proficiency, and 110 as high proficiency; the corresponding counts for males were 503, 559, and 78.

We also conducted an additional analysis by race. Because the original race/ethnicity variable included several categories with relatively small sample sizes, we coded it dichotomously as White and non-White for this comparison. The analytic sample included 852 White students (39.32%) and 1315 non-White students (60.68%). Among White students, 258 were classified as low proficiency, 468 as medium proficiency, and 126 as high proficiency; the corresponding counts for non-White students were 595, 658, and 62. Additional details regarding data processing and descriptive statistics are provided in the project's GitHub repository.[3]

2.2.2. AES models

To illustrate our approach, we chose four AES models that use students' written responses to predict essay scores and represent different paradigms: a traditional statistical model, a deep learning model, and two configurations based on ChatGPT. Specifically, these AES models included (1) an ordinal logistic regression model with term frequency–inverse document frequency (TF–IDF) features, (2) a fine-tuned Bidirectional Encoder Representations from Transformers (BERT) model, (3) a ChatGPT zero-shot model, and (4) a ChatGPT few-shot model.

The writing samples were randomly divided into two sets: 80% for training and 20% for testing. The ordinal logistic regression model and the BERT model were trained on the training data, with hyperparameters optimized through grid search. In contrast, the ChatGPT models in the present study were not trained or fine-tuned on the dataset. Instead, ChatGPT was guided to assign scores based on a few example essays provided in the prompt, a procedure commonly referred to as few-shot prompting. For comparison, zero-shot prompting asked the model to perform the same scoring task without providing any examples.

Predictive bias was evaluated by sex and race. For the sex-based analysis, female students were treated as the focal group and male students as the reference group. The race-based analysis was included as an additional empirical demonstration and is reported in Section 3.4 and Appendix C. All four AES models were evaluated on the same test set to ensure comparability of both predictive performance and bias detection. All model configurations, parameter settings, and implementation details can be found in the same project repository.

2.2.2.1. Ordinal logistic regression with TF–IDF features. TF–IDF was used to extract textual features from student essays, capturing word-frequency patterns adjusted for their relative importance across the corpus. These features were then entered into an ordinal logistic regression model to predict proficiency levels. This AES model represents a traditional statistical approach that relies on lexical features rather than contextual representations of text. The model was trained on the training set, and hyperparameters were selected through grid search. The top 30 TF–IDF features from the final model are reported in Appendix B.

2.2.2.2. Fine-tuned BERT model. BERT is a deep learning architecture based on transformer networks that captures bidirectional contextual information from text. In this AES model, a pretrained BERT-base model was fine-tuned to classify essays into three proficiency levels. The model was trained on the training set, with hyperparameters selected using a validation split from the training data. The tuned hyperparameters included learning rate, batch size, number of training epochs, and maximum sequence length. This approach represents a modern neural network that directly learns semantic and syntactic relationships from full essay texts.

2.2.2.3. ChatGPT-based models. The third and fourth AES models used ChatGPT-4o-mini, implemented through the OpenAI API, in two prompting configurations: zero-shot and few-shot. In the zero-shot setting, ChatGPT performed the scoring task directly from the instruction without seeing any examples. In the few-shot setting, the prompt included several example essays and their scores to guide the model's scoring decisions. Hyperparameter tuning was not required; instead, prompt design served as the primary configuration step. Prompts were constructed based on the holistic scoring rubric, and chain-of-thought prompting techniques were explored to improve scoring consistency. ChatGPT-4o-mini was selected for its balance of speed, cost efficiency, and performance. Details of the prompts used in both configurations are provided in Appendix A.

2.2.3. Evaluation of AES performance

The quadratic-weighted kappa (QWK) is a common and preferred index of AES performance. In this study, QWK was used to

[2] The full prompt text for the Facial Action Coding System task is available in the public PERSUADE 2.0 release (https://www.kaggle.com/datasets/nbroad/persaude-corpus-2)

[3] Project repository: https://github.com/Guanyu0001/Assessing-Writing-Project

evaluate agreement between the AES-predicted three-category score and the three-category score recoded from human ratings. The ordinal logistic regression with TF–IDF features achieved a QWK of 0.73, and the fine-tuned BERT model achieved a QWK of 0.80. The ChatGPT zero-shot model and the ChatGPT few-shot model showed lower agreement (QWK = 0.40 and 0.51, respectively). Under conventional kappa interpretation guidelines, values from 0.21 to 0.40 indicate fair agreement, values from 0.41 to 0.60 indicate moderate agreement, and values from 0.61 to 0.80 indicate substantial agreement (Landis & Koch, 1977). In AES practice, ETS has also treated QWK values around 0.70 as acceptable for operational use (Wang & von Davier, 2014). Accordingly, only the TF–IDF and BERT models exceeded this operational benchmark, whereas the ChatGPT zero-shot and few-shot models showed fair-to-moderate agreement for the recoded three-category scores. These results are consistent with prior studies using the same dataset and ChatGPT-4o-mini, where zero-shot performance ranged from 0.25 to 0.50 and few-shot performance from 0.33 to 0.60 (Plasencia-Calaña, 2025). Although the primary aim of the present study is not to optimize AES performance, the four models provide a realistic range of scoring performance for illustrating the proposed predictive bias measures.

3. Results

3.1. Conditional disparity ratio

Table 1 presents the unconditional and conditional disparity ratios across predicted score categories for the four AES models. For example, in the ordinal logistic regression model for predicting writing samples as level 3 ($\hat{Y} = 3$), the unconditional model suggests almost no group difference (1.049), i.e., no predictive bias, whereas the conditional model indicates potential bias in favor of female students (1.559) at the medium proficiency level ($\theta = 2$) and potential bias against female students (0.595) at the high proficiency level ($\theta = 3$). The inconsistent conclusion about bias between unconditional and conditional results is also evident for predictions at the other proficiency levels ($\hat{Y} = 1$ or 2), emphasizing that without accounting for proficiency, conclusions about potential predictive bias can be incomplete or misleading. The inconsistency in results between conditional and unconditional disparity ratios can also be seen in the other three AES models, as illustrated in Table 1. These differences across AES models are expected because the models differ in their text representations, training procedures, calibration, and resulting score distributions. The prompt-based ChatGPT models, in particular, represent a different scoring setup because they rely on rubric instructions and examples in the prompt rather than task-specific parameter training.

It is noteworthy that several conditional disparity ratios shown in Table 1 are not applicable because they could not be meaningfully computed. Specifically, the off-diagonal cells, i.e., cases such as $\theta = 1$ with $\hat{Y} = 3$ and $\theta = 3$ with $\hat{Y} = 1$, could not be calculated because one of the groups had no essays predicted at those score levels. This pattern indicates that the AES models generally produced accurate predictions close to the true proficiency levels and rarely misclassified essays across distant proficiency levels, i.e., predicting low-proficiency essays as high, or vice versa.

To facilitate interpretation of the resulting 3 × 3 matrix of conditional disparity ratios across proficiency levels (θ =1, 2, and 3) and predicted score categories ($\hat{Y}$ =1,2,3), we provide a heatmap visualization that summarizes how the direction and magnitude of disparity vary across proficiency levels and predicted score categories (see Fig. 1).

Table 1
Disparity Ratio with or without Conditioning on Writing Proficiency Level for Four AES Models.

AES Models	Unconditional	*Conditioning on Writing Proficiency Level*		
		$\theta = 1$	$\theta = 2$	$\theta = 3$
Ordinal Logistic Regression				
$\hat{Y} = 1$	0.833	1.060	0.845	—
$\hat{Y} = 2$	1.116	0.765	0.995	1.944
$\hat{Y} = 3$	1.049	—	1.559	0.595
BERT				
$\hat{Y} = 1$	0.757	0.930	1.462	—
$\hat{Y} = 2$	1.057	1.292	0.966	0.521
$\hat{Y} = 3$	1.771	—	1.624	1.319
ChatGPT Zero-Shot				
$\hat{Y} = 1$	0.815	0.904	0.863	0.556
$\hat{Y} = 2$	1.464	3.875	1.213	1.078
$\hat{Y} = 3$	—	—	—	—
ChatGPT Few-Shot				
$\hat{Y} = 1$	0.741	0.797	0.875	—
$\hat{Y} = 2$	1.295	2.510	1.064	0.965
$\hat{Y} = 3$	1.107	—	—	0.833

Note. Em dashes (—) indicate cases in which the disparity ratio could not be meaningfully computed because one of the groups had no predicted essays at that score level. The symbol θ denotes the true writing proficiency level, using the human rating score as a proxy (1 = low, 2 = medium, 3 = high). Values in the "Unconditional" column were computed across all proficiency levels, whereas conditional values were computed separately for each level of θ. Female students were treated as the focal group and male students as the reference group.

3.2. Conditional disparity difference

Because ratio-based measures may become unstable when probabilities are very small, we also report the conditional disparity difference, defined as the difference in predicted probabilities between groups at the same proficiency level. As shown in Table 2, the conditional disparity differences, like the conditional disparity ratios, show that conclusions about bias change after proficiency is taken into account. Using the ordinal logistic regression model as an example, the unconditional disparity difference for a predicted score level of 3 was 0.004, indicating nearly equal predicted probabilities and minimal bias. After proficiency is taken into account, the results show that female students have a slight advantage (0.024) at the medium proficiency level but experience greater bias (−0.283) at the high proficiency level.

To facilitate interpretation of the corresponding 3 × 3 conditional disparity difference matrix across proficiency levels (θ =1,2,3) and predicted score categories ($\hat{Y}$ =1,2,3), we provide a heatmap visualization that summarizes how group differences vary across proficiency levels and predicted score categories (see Fig. 2).

3.3. Comparison with the regression-based approach

To further illustrate the differences between the proposed measures and the regression-based approach described by Loukina et al. (2019), we also computed the corresponding OSA, OSD, and CSD values. The results are presented in Table 3. Overall, the regression-based indices were uniformly small across all four AES models. In particular, the OSA and OSD values were all close to zero, suggesting little detectable group difference under these global regression summaries. Although Loukina et al. (2019) further proposed evaluating CSD using an analysis of variance test for the difference in R^2 (ΔR^2), none of the CSD results shown here met the $\alpha = 0.01$ threshold used in Loukina et al.'s example.

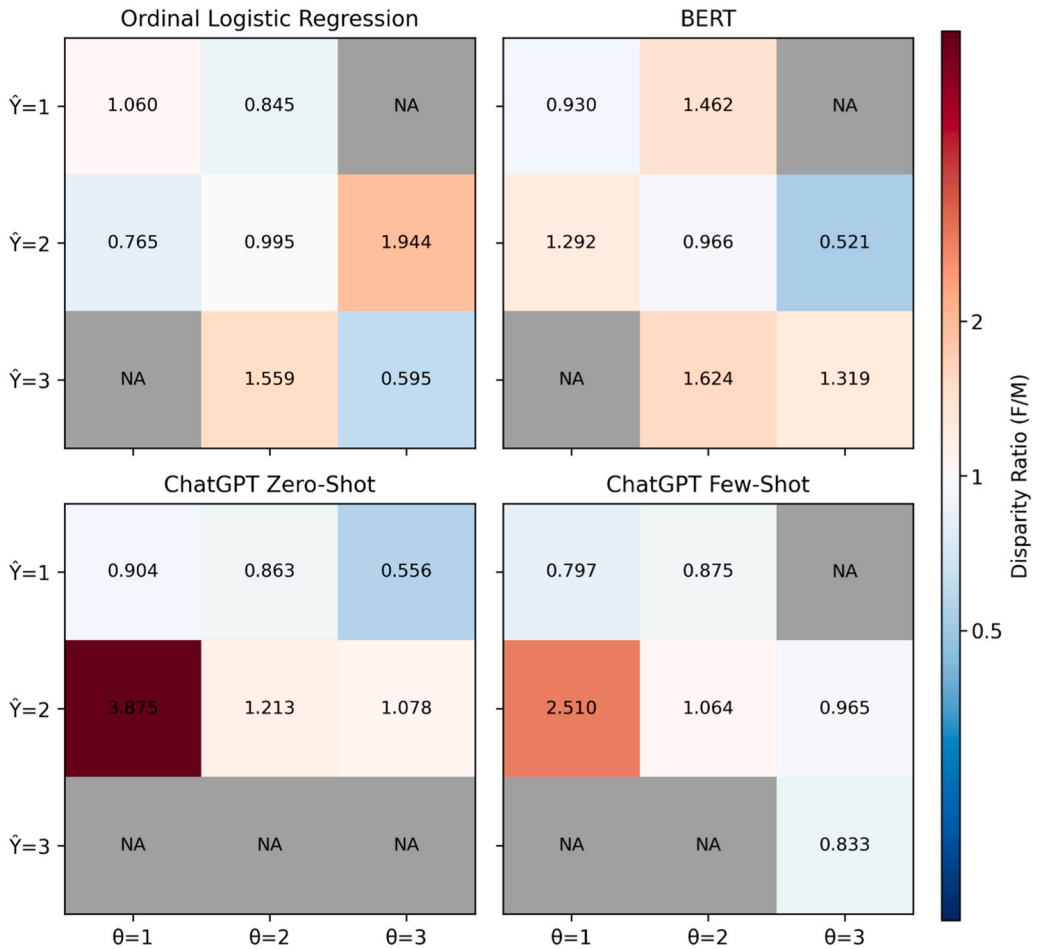


Fig. 1. Conditional Disparity Ratio Heatmaps Across Models*Note*. Each panel shows the 3 × 3 conditional disparity ratio (F/M) across proficiency levels (columns) and predicted score categories (rows). Cell labels report the conditional disparity ratio values; NA cells are shown in gray. Colors are centered at 1.0 (no disparity in ratio), and the color scale is reciprocal-symmetric so that ratios r and $1/r$ have equal intensity (e.g., 0.5 and 2.0 indicate the same magnitude in opposite directions).

Table 2
Disparity Difference with or without Conditioning on Writing Proficiency Level for Four AES Models.

AES Models	Unconditional	*Conditioning on Writing Proficiency Level*		
		$\theta = 1$	$\theta = 2$	$\theta = 3$
Ordinal Logistic Regression				
$\widehat{Y} = 1$	-0.065	0.048	-0.020	0.000
$\widehat{Y} = 2$	0.061	-0.048	-0.004	0.283
$\widehat{Y} = 3$	0.004	0.000	0.024	-0.283
BERT				
$\widehat{Y} = 1$	-0.084	-0.056	0.016	0.000
$\widehat{Y} = 2$	0.034	0.056	-0.032	-0.192
$\widehat{Y} = 3$	0.051	0.000	0.016	0.192
ChatGPT Zero-Shot				
$\widehat{Y} = 1$	-0.132	-0.093	-0.083	-0.067
$\widehat{Y} = 2$	0.132	0.093	0.083	0.067
$\widehat{Y} = 3$	0.000	0.000	0.000	0.000
ChatGPT Few-Shot				
$\widehat{Y} = 1$	-0.137	-0.179	-0.043	0.042
$\widehat{Y} = 2$	0.137	0.179	0.043	-0.033
$\widehat{Y} = 3$	0.000	0.000	0.000	-0.008

Note. The symbol θ denotes the true writing proficiency level, using the human rating score as a proxy (1 = low, 2 = medium, 3 = high). Values in the "Unconditional" column were computed across all proficiency levels, whereas conditional values were computed separately for each level of θ. Female students were treated as the focal group and male students as the reference group.

These findings contrast with the results from our proposed methods, the conditional disparity ratio and conditional disparity difference shown in Tables 1 and 2, which revealed more differentiated subgroup patterns across predicted score categories and proficiency levels. This contrast is consistent with the argument presented earlier: because the regression-based approach reduces fairness to a single global summary, it may be relatively insensitive to category-specific or proficiency-specific disparity patterns in ordinal AES outcomes.

3.4. Additional analysis by race groups

To examine whether the overall pattern of findings extended to a different subgroup definition, we conducted an additional analysis using a dichotomized race variable (White vs. non-White). Applying the same analytic framework, the conditional disparity ratio and conditional disparity difference again revealed subgroup predictive biases that differed across predicted score categories and proficiency levels, whereas the regression-based baseline indices, including OSA, OSD, and CSD, produced comparatively small values. Overall, the race-based analysis yielded the same general conclusion as the sex-based analysis, although the specific values differed across models and score categories. The dichotomized race variable was used for illustration and should not be interpreted as a recommended way to operationalize race in fairness evaluation. When multiple racial groups need to be evaluated, the same procedure can be applied to planned group comparisons, such as comparing a selected focal group with a designated reference group based on the purpose of fairness evaluation. Detailed race-based tables and heatmap visualizations are provided in Appendix C.

4. Discussion

This study proposed a framework for detecting potential predictive bias in AES and compared it with existing approaches in an empirical example. Specifically, the framework is based on the conditional disparity ratio and conditional disparity difference. Compared with unconditional measures and regression-based approaches, these conditional measures provide more meaningful comparisons because they evaluate potential bias among students at the same proficiency level. This approach not only allows researchers to separate differences in writing proficiency from prediction bias but also makes it possible to identify patterns of bias that vary across proficiency levels. For example, an AES model may underestimate low-proficiency students while simultaneously overestimating high-proficiency students, a pattern that unconditional measures and regression-based methods would obscure. By addressing these limitations, the conditional disparity ratio and difference offer a more precise and theoretically grounded framework for examining fairness in AES.

4.1. AES prediction accuracy: A prerequisite for fairness evaluation

It is important to recognize that Tables 1 and 2 indicate potential bias, not AES model accuracy. A model may appear "fair" by these measures but still perform poorly, making predictive bias results meaningless without adequate predictive accuracy. For example, the ChatGPT zero-shot model shows no group difference for the highest proficiency level simply because it rarely assigned high scores. In this case, the disparity difference shows no bias, not because the model was fair, but because it fails to generate valid predictions. This example underscores that fairness measures are only meaningful when applied to models with acceptable accuracy. It also highlights a

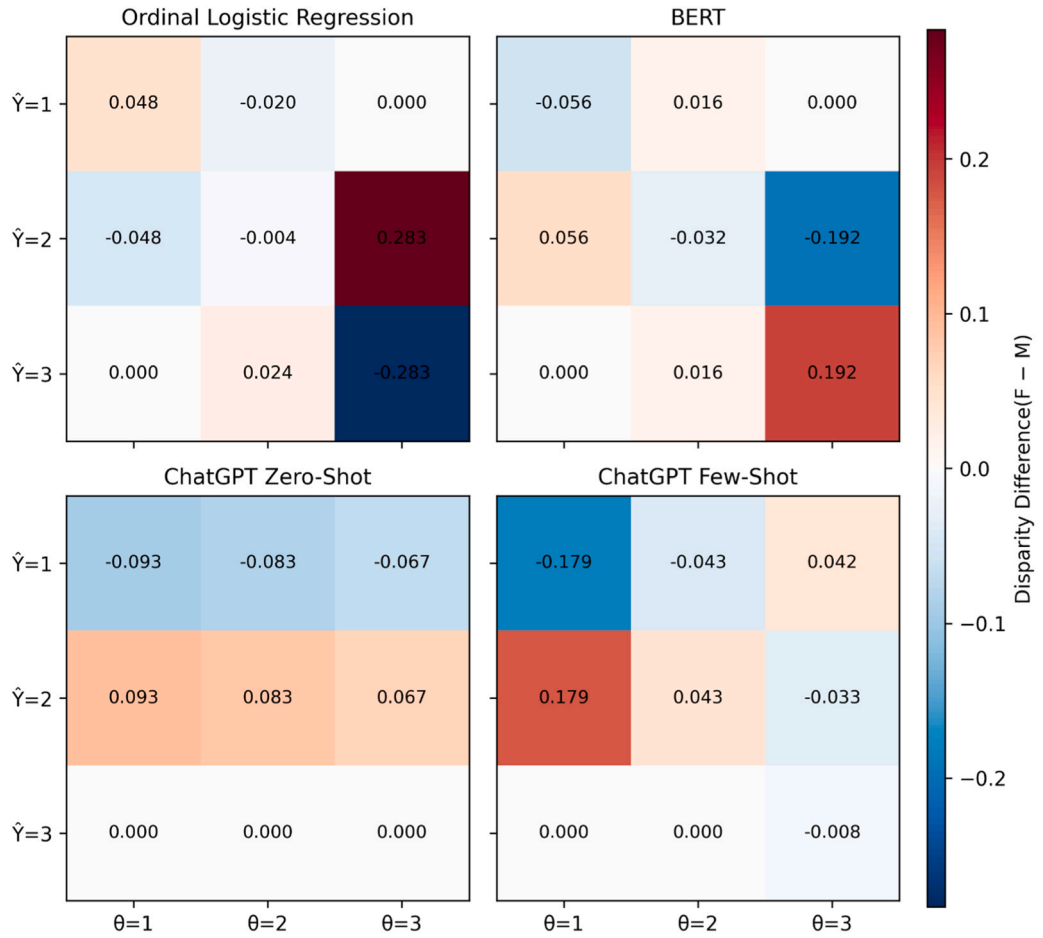


Fig. 2. *Conditional Disparity Difference Heatmaps Across Models Note*. Each panel shows the 3 × 3 conditional disparity difference (F − M) across proficiency levels (columns) and predicted score categories (rows). Cell labels report the original difference values; NA cells are shown in gray. Colors are centered at 0 (no disparity), with equal color intensity for differences of the same absolute magnitude in opposite directions (e.g., −0.20 and 0.20).

Table 3
Regression-Based Approach Results Across AES Models.

			CSD	
AES Model	OSA	OSD	ΔR^2	p
Ordinal Logistic Regression	0.001	0.004	0.001	0.621
BERT	0.000	0.000	0.003	0.237
ChatGPT Zero-Shot	0.001	0.000	0.006	0.034
ChatGPT Few-Shot	0.007	0.000	0.006	0.043

Note. OSA = overall score accuracy; OSD = overall score difference; CSD = conditional score difference. These indices were implemented based on the method description in Section 4.2 of Loukina et al. (2019). The reported p-values for CSD follow the analysis of variance test for the nested models proposed by Loukina et al. (2019), using $\alpha = 0.01$ as the significance criterion.

broader limitation of using generic large language models for AES: while zero-shot LLMs can perform a wide range of linguistic and reasoning tasks, they often produce unreliable results for task-specific writing evaluation (Lee et al., 2024; Liu et al., 2025; Q. Wang & Gayed, 2024). Without careful prompt design and task-specific adaptation, their predictions lack the accuracy required for valid educational assessment. As emphasized by Xue et al. (2025), developing a dependable AES based on LLMs requires deliberate prompt construction, fine-tuning, and systematic validation.

4.2. Proxy for true writing proficiency

As previously noted, because students' true writing proficiency is unobservable, a proxy is required for conditioning in our predictive bias analyses. In our demonstration, we used human ratings from the same task under evaluation. This choice does not imply that it is the only valid option. When more robust indicators are available, such as scores from other validated writing assessments or students' coursework performance, they can also serve as effective proxies, and the proposed methodology remains applicable regardless of the proxy selected.

In practice, however, most publicly available writing datasets include only human ratings for the same task under fairness evaluation, making them one of the most practical choices. Prior research in AES has similarly shown that subgroup bias may appear minimal in the full sample but becomes more pronounced when comparisons are conditioned on human scores (Lottridge & Young, 2022). This pattern aligns with the central argument of our methodology: even in the absence of a strong external criterion, conditioning on human ratings as a proxy for proficiency can yield more meaningful evidence of fairness than unconditional group comparisons.

4.3. Scale of writing score is relevant for detecting predictive bias

A key issue in fairness research for AES is that outcome variables are often reported as ordinal proficiency levels (e.g., 1–6 in PERSUADE, 0–5 in TOEFL, 1–6 in GRE). However, many commonly used fairness methods treat these scores as interval data and apply ordinary least squares regression, implicitly assuming equal distances between adjacent levels. This assumption may not hold in practice, as the steps of progressing from one level to the next are unlikely to be uniform. This consideration also supports our use of the conditional disparity ratio and conditional disparity difference, which compare probabilities across ordered score categories rather than treating the outcome as interval.

Even when outcome variables are continuous, they can be meaningfully grouped into ordinal levels using well-justified cutoffs to facilitate interpretation and reporting, as demonstrated in large-scale assessments such as the Progress in International Reading Literacy Study (PIRLS). This flexibility makes the proposed approach well suited for analyzing AES outcomes. In addition, the probability-based formulation for ordinal outcomes also leaves room for extensions that reflect the severity of disagreement between observed and predicted score categories, so that larger mismatches can be weighted more heavily than smaller ones.

4.4. Expanding applications of our bias detection methodology

Our predictive bias detection framework is intended for high-stakes educational settings beyond automated essay scoring, where automated or semi-automated scoring increasingly informs consequential decisions. Conceptually, our predictive bias framework is aligned with long-standing psychometric practice used in DIF analyses, in that both rely on conditional comparisons between focal and reference groups given a proficiency proxy. In settings where whole-test information and appropriate matching variables are available (e.g., total score or rest score), DIF-style analyses can also be applied to examine whether observed scores or rating outcomes show differential functioning across groups, providing a complementary diagnostic perspective grounded in established educational and psychological measurement practice.

However, we emphasize predictive bias as a distinct and practically important challenge in AES because the object of evaluation is a scoring algorithm and the key inferential requirement is generalization to unseen responses (e.g., held-out testing or cross-validation). In many AES applications, the available data are limited to single-task responses and associated human scores, rather than the broader test structure that supports classic matching variables; consequently, strong proficiency proxies may be unavailable and "true" performance is only imperfectly observed through human ratings. These constraints make fairness evaluation in AES both more model-dependent and more sensitive to data limitations than traditional DIF workflows, reinforcing the need for an explicitly predictive, out-of-sample evaluation framework.

4.5. Practical implementation

In practice, fairness should be monitored regularly. Evaluating predictive bias requires at least a small audit sample of essays that are scored by both the AES system and human raters. Without human scores on new essays, it is not possible to evaluate whether the system's predictions remain accurate and fair over time. This is consistent with high-stakes AES practice, in which human scoring is retained as an anchor while automated scoring is used to support, rather than replace, human judgment (Ramineni et al., 2014). A practical strategy is to collect a small, representative sample of human-scored essays in each new administration and use this labeled subset to recompute fairness metrics and check whether predictive bias changes over time. Essays without human scores may still be useful for monitoring broader shifts in response patterns, but they cannot replace human-scored samples for fairness evaluation.

4.6. Limitations

The proposed measures have several limitations that should be considered when interpreting the findings. First, the present study used human ratings from the same writing task as the proficiency proxy because alternative, reliable proficiency measures were not available. Although this is a practical choice for many publicly available AES datasets, single-task human ratings may not fully represent students' true writing proficiency. Second, conditional analyses require dividing the sample by both proficiency level and

subgroup membership. This can reduce the number of cases in each comparison, increase uncertainty, and make some conditional disparity ratios or differences difficult to compute when cells are sparse or empty. Finally, the proposed measures identify evidence of potential predictive bias rather than conclusive bias. When such evidence is found, the potential source of bias should be further investigated in relation to the construct being assessed, the intended use of the scores, and the available validity evidence, with input from relevant stakeholders and content experts when appropriate.

5. Future directions

While detecting bias is a crucial step, an equally important consideration is whether such bias can be effectively mitigated. Although this study did not directly address mitigation within its permissible scope, future research can explore several key challenges in this area, particularly across the three broad approaches to bias mitigation in machine learning: data pre-processing, in-model processing, and outcome post-processing (Li et al., 2023; Pessach & Shmueli, 2022).

Bias reduction during pre-processing adjusts training data before training to limit input bias, such as reweighting samples or balancing groups. These methods are practical and relatively easy to implement, but their effectiveness cannot be guaranteed because modifications to the data do not always produce fairer models. In AES settings, an especially relevant pre-processing concern is that human ratings are often used as training labels and may contain rater effects. When feasible (e.g., with multi-rater designs), rater-mediated measurement approaches such as many-facet Rasch measurement can be used to examine and adjust for rater effects, alongside operational practices such as rater calibration and adjudication (Myford & Wolfe, 2003; Myford & Wolfe, 2004). Second, fairness can also be addressed during model training. In-model approaches incorporate fairness constraints or adversarial objectives to reduce bias in the estimated parameters. Although this strategy directly targets bias within the model, it is technically complex, computationally demanding, and may be difficult for applied users to implement. Finally, fairness can be addressed at the post-processing stage by adjusting predicted scores after model estimation to equalize outcomes. While such methods are relatively straightforward to apply and can achieve desired fairness criteria, they raise concerns about validity and transparency because they modify scores after the scoring process has been completed.

Future research should move beyond general discussions of mitigation to focus on the unique context of AES. As emphasized earlier, the pursuit of fairness is inevitable but can be controversial. When AES operates in high-stakes educational settings, mitigation strategies must ensure not only statistical fairness but also educational values and consequences. Comprehensive reviews are needed to evaluate which mitigation strategies are most appropriate for AES and how their effectiveness varies across model types. Reviews should thoroughly assess old and new developments not only in machine learning and LLMs, but also in psychometrics to identify the most suitable and effective way of mitigating bias in AES. For example, the emergence of LLMs warrants special attention because their fairness depends not only on training data but also on how prompts are designed, how example responses are chosen, and how the models are tuned for specific tasks.

In closing, there remains an urgent need for comprehensive and actionable guidelines to support bias detection and mitigation in AES. This includes clear recommendations for best practices in data collection, thoughtful evaluation of AES design, and transparent reporting standards. Our study contributes to the development of a robust methodology for detecting predictive bias.

CRediT authorship contribution statement

Guanyu Chen: Writing – original draft, Methodology, Formal analysis, Conceptualization. **Amery D. Wu:** Writing – review & editing, Supervision, Conceptualization. **Chiyu Zhang:** Software, Formal analysis.

Disclosure statement

The authors report there are no competing interests to declare.

Appendix A

Prompt Design for Zero-Shot and Few-Shot Analysis

In this study, we considered how to design prompts for ChatGPT when applying it to essay scoring. Because each word or instruction in the prompt contributes to the token count, it is impractical to provide overly long or detailed scoring guides for every response. Instead, our approach was to summarize the essential features of the holistic rubric and present them in a concise but meaningful way. The original six-level holistic scoring rubric (see https://github.com/scrosseye/persuade_corpus_2.0/blob/main/sat_rubric_only_source_based.pdf)was collapsed into three proficiency levels (1, 2, and 3) to reduce complexity while retaining interpretability. The key idea was to preserve as much meaning as possible in a compressed form, consistent with measurement principles that emphasize clarity and parsimony.

In designing the prompts, we drew on research showing that large language models often benefit from structured instructions and step-by-step reasoning, sometimes referred to as chain-of-thought prompting.[4] For the zero-shot condition, the model was explicitly

[4] This prompt design was informed by the official OpenAI prompting guide, which notes that prompting the model to think step by step can be useful in some settings: https://developers.openai.com/cookbook/examples/gpt4-1_prompting_guide

instructed to follow the rubric, think through the essay in stages, and then return only the final holistic score. To reduce the risk of irrelevant or verbose responses, the output was restricted to the integer score using a function-call format. The full system message for the zero-shot condition was reproduced below:

```
You are an essay rater following the Rubric.
Think step by step before deciding, and return the final score.
Rubric:
3: An essay in this category demonstrates clear and consistent mastery. It effectively develops a point of view with strong and accurate use of source evidence. The essay is well organized and focused, showing coherence and smooth progression of ideas. Language use is precise, varied, and appropriate, with meaningful sentence variety. Only minor or occasional errors may appear, but they do not interfere with meaning.
2: An essay in this category demonstrates adequate but uneven command. It develops a point of view and shows some critical thinking, though examples and evidence may be limited or inconsistently used. Organization is generally clear, but lapses in focus or coherence are present. Language is ordinary or inconsistent, sometimes weak or inappropriate in word choice, and sentence variety may be limited. Noticeable errors in grammar, usage, or mechanics occur, though meaning is still understandable overall.
1: An essay in this category demonstrates very limited or no mastery. The point of view may be vague, seriously limited, or entirely absent, with weak, irrelevant, or no use of evidence. The essay is disorganized or incoherent, with poor focus and progression of ideas. Vocabulary is very limited or incorrect, sentence structure is severely flawed, and pervasive errors in grammar, usage, or mechanics interfere with meaning.
```

For the few-shot condition, we extended this structure by embedding six human-scored essays as examples from the training set, with two drawn from each of the three collapsed proficiency levels. In practice, one essay was drawn from each of the paired original scores (1 and 2 mapped to proficiency level 1, 3 and 4 mapped to proficiency level 2, and 5 and 6 mapped to proficiency level 3). These examples were presented as alternating user–assistant pairs, where the essay text was shown as user input and the correct score as the assistant's output, consistent with OpenAI's few-shot prompting guidelines. Because the full prompt text with six embedded essays is lengthy, the complete few-shot setup was made available in the same project repository.

Across both conditions, the system message was kept consistent: the model was positioned as an essay rater, guided by the rubric, and instructed to think step by step before deciding. This balance between conciseness and structured reasoning reflects insights from recent work on prompt engineering in educational measurement. Xue et al. (2025) emphasize that prompts benefit from three strategies: providing information, including examples, and encouraging stepwise thought. Our design integrated all three strategies but tailored them to the constraints of AES.

Appendix B

Top 30 TF–IDF Features Used in the Ordinal Logistic Regression

The TF–IDF feature space is large; for transparency, we report here the Top 30 TF–IDF features ranked by the absolute value of the fitted regression coefficients from the final ordinal logistic regression model.

Feature	Coefficient	Absolute value of coefficient
could	237.6975	237.6975
this	220.5549	220.5549
more	188.5148	188.5148
facs	186.2356	186.2356
can	170.2843	170.2843
software	166.7276	166.7276
school	164.718	164.718
the computer	163.7264	163.7264
system	158.9163	158.9163
being	158.8486	158.8486
student	158.0236	158.0236
students	154.9157	154.9157
not	152.7399	152.7399
privacy	152.4241	152.4241
would	148.1939	148.1939
with	146.2907	146.2907
are	144.8747	144.8747
these	144.377	144.377
if	142.8776	142.8776
app	141.4262	141.4262
by	139.8737	139.8737

(*continued on next page*)

(*continued*)

Feature	Coefficient	Absolute value of coefficient
is not	137.7847	137.7847
kid	131.7957	131.7957
their	130.7313	130.7313
computer	128.5199	128.5199
alto	127.1582	127.1582
the system	125.8741	125.8741
technology	121.4798	121.4798
may	114.2862	114.2862
classroom	113.1923	113.1923

Note. For consistency across AES models, we used the same minimally processed raw text input and did not apply stopword removal or stemming/lemmatization. Function words and surface-form choices can carry information about linguistic style and grammatical proficiency, which is relevant for AES and differs from topic-modeling settings where such tokens are often removed.

Appendix C

Additional Analysis by Race

C.1 Conditional Disparity Ratio

Table C1

Disparity Ratio with or without Conditioning on Writing Proficiency Level for Four AES Models

AES Models	Unconditional	*Conditioning on Writing Proficiency Level*		
		$\theta = 1$	$\theta = 2$	$\theta = 3$
Ordinal Logistic Regression				
$\widehat{Y} = 1$	0.603	0.925	0.382	—
$\widehat{Y} = 2$	1.255	1.385	1.138	1.677
$\widehat{Y} = 3$	1.644	—	0.876	0.699
BERT				
$\widehat{Y} = 1$	0.691	0.946	0.935	—
$\widehat{Y} = 2$	1.028	1.215	0.972	1.398
$\widehat{Y} = 3$	2.796	—	2.337	0.881
ChatGPT Zero-Shot				
$\widehat{Y} = 1$	0.740	1.053	0.701	0.629
$\widehat{Y} = 2$	1.706	0.436	1.548	1.067
$\widehat{Y} = 3$	—	—	—	—
ChatGPT Few-Shot				
$\widehat{Y} = 1$	0.663	0.902	0.673	—
$\widehat{Y} = 2$	1.374	1.501	1.190	0.903
$\widehat{Y} = 3$	—	—	—	—

Note. Em dashes (—) indicate cases in which the disparity ratio could not be meaningfully computed because one group had no predicted essays at that score level. The symbol θ denotes the true writing proficiency level, with the human rating used as a proxy (1 = low, 2 = medium, 3 = high). Values in the Unconditional column were computed across all proficiency levels, whereas conditional values were computed separately within each level of θ. For these analyses, the disparity ratio was defined as the predicted probability for White students divided by the predicted probability for non-White students.

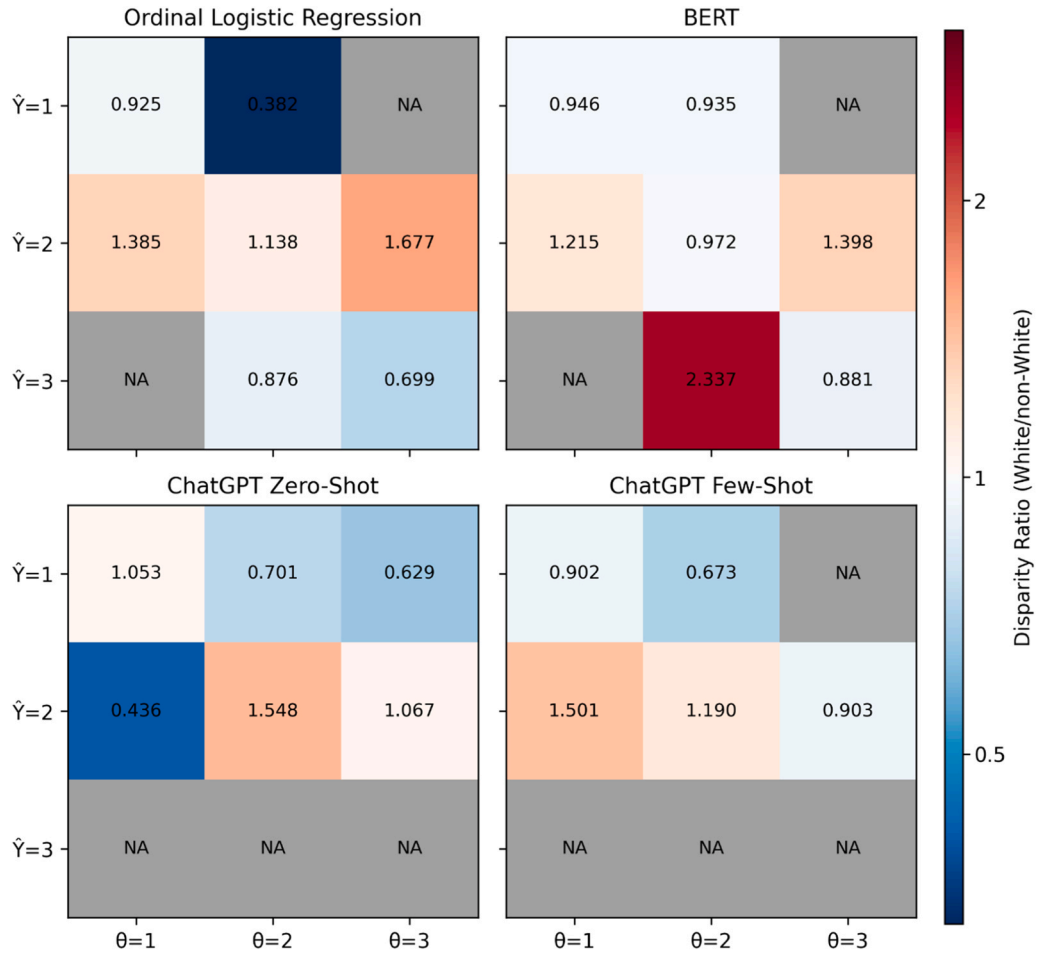


Figure C1. *Conditional Disparity Ratio Heatmaps Across Models. Note.* Each panel shows the 3 × 3 conditional disparity ratio (White/non-White) across proficiency levels (columns) and predicted score categories (rows). Cell labels report the ratio values; NA cells are shown in gray. Colors are centered at 1.0 (no disparity in ratio), and the color scale is reciprocal-symmetric so that ratios r and $1/r$ have equal intensity (e.g., 0.5 and 2.0 indicate the same magnitude in opposite directions)

C.2 Conditional Disparity Difference

Table C2

Disparity Difference with or without Conditioning on Writing Proficiency Level for Four AES Models

AES Models	Unconditional	*Conditioning on Writing Proficiency Level*		
		$\theta = 1$	$\theta = 2$	$\theta = 3$
Ordinal Logistic Regression				
$\hat{Y} = 1$	-0.171	-0.063	-0.100	0.000
$\hat{Y} = 2$	0.128	0.063	0.107	0.208
$\hat{Y} = 3$	0.043	0.000	-0.007	-0.208
BERT				
$\hat{Y} = 1$	-0.109	-0.043	-0.003	0.000
$\hat{Y} = 2$	0.016	0.043	-0.027	0.092
$\hat{Y} = 3$	0.092	0.000	0.029	-0.092
ChatGPT Zero-Shot				
$\hat{Y} = 1$	-0.190	0.049	-0.193	-0.057
$\hat{Y} = 2$	0.190	-0.049	0.193	0.057
$\hat{Y} = 3$	0.000	0.000	0.000	0.000
ChatGPT Few-Shot				
$\hat{Y} = 1$	-0.182	-0.082	-0.120	0.032
$\hat{Y} = 2$	0.171	0.082	0.120	-0.097
$\hat{Y} = 3$	0.011	0.000	0.000	0.065

Note. The symbol θ denotes the true writing proficiency level, with the human rating used as a proxy (1 = low, 2 = medium, 3 = high). Values in the Unconditional column were computed across all proficiency levels, whereas conditional values were computed separately within each level of θ. For these analyses, the disparity difference was defined as the predicted probability for White students minus the predicted probability for non-White students.

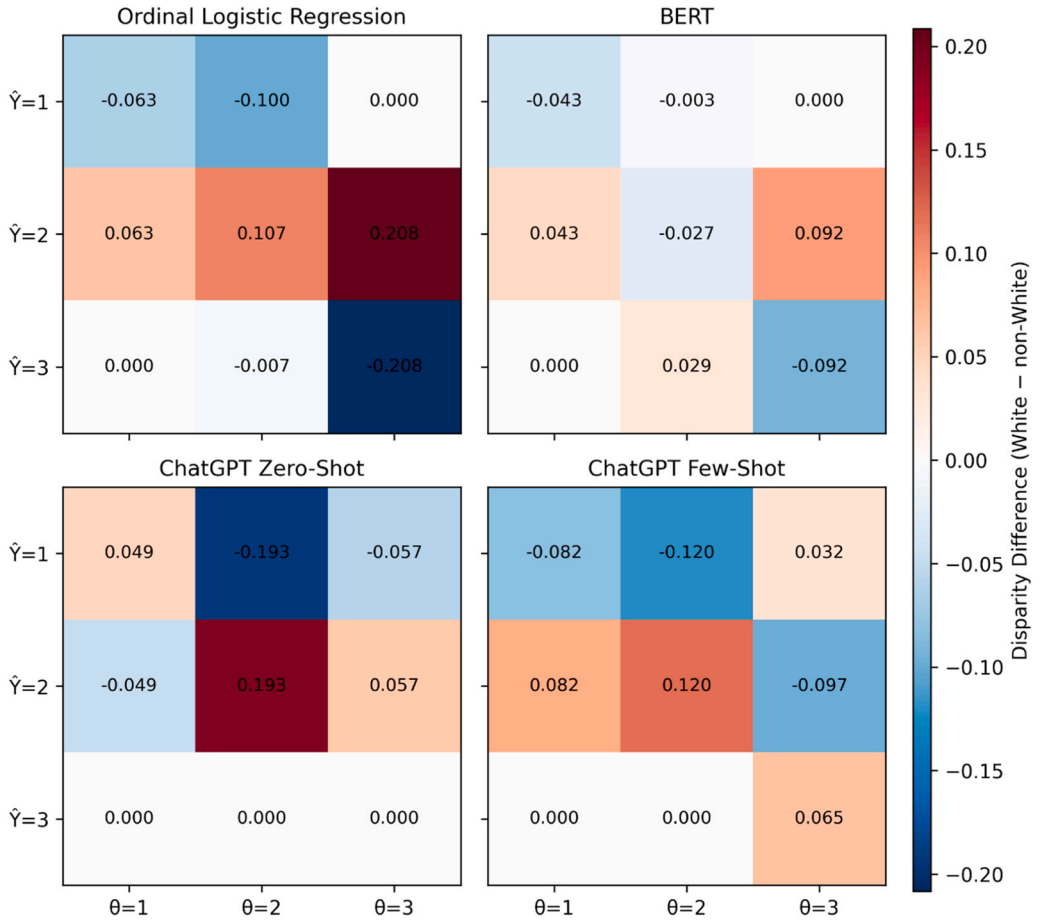


Figure C2. *Conditional Disparity Difference Heatmaps Across Models. Note.* Each panel shows the 3 × 3 conditional disparity difference (White − non-White) across proficiency levels (columns) and predicted score categories (rows). Cell labels report the difference values; NA cells are shown in gray. Colors are centered at 0 (no disparity), with equal color intensity for differences of the same absolute magnitude in opposite directions (e.g., −0.20 and 0.20)

C.3 Comparison with the Regression-Based Approach

Table C3

Regression-Based Approach Results Across AES Models

			CSD	
AES Model	OSA	OSD	ΔR^2	p
Ordinal Logistic Regression	0.000	0.001	0.003	0.244
BERT	0.004	0.002	0.001	0.531
ChatGPT Zero-Shot	0.000	0.002	0.008	0.014
ChatGPT Few-Shot	0.004	0.002	0.006	0.030

Note. OSA = overall score accuracy; OSD = overall score difference; CSD = conditional score difference. These indices were implemented based on the method description in Section 4.2 of Loukina et al. (2019). The reported p-values for CSD follow the analysis of variance test for the nested models proposed by Loukina et al. (2019), using $\alpha = 0.01$ as the significance criterion.

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